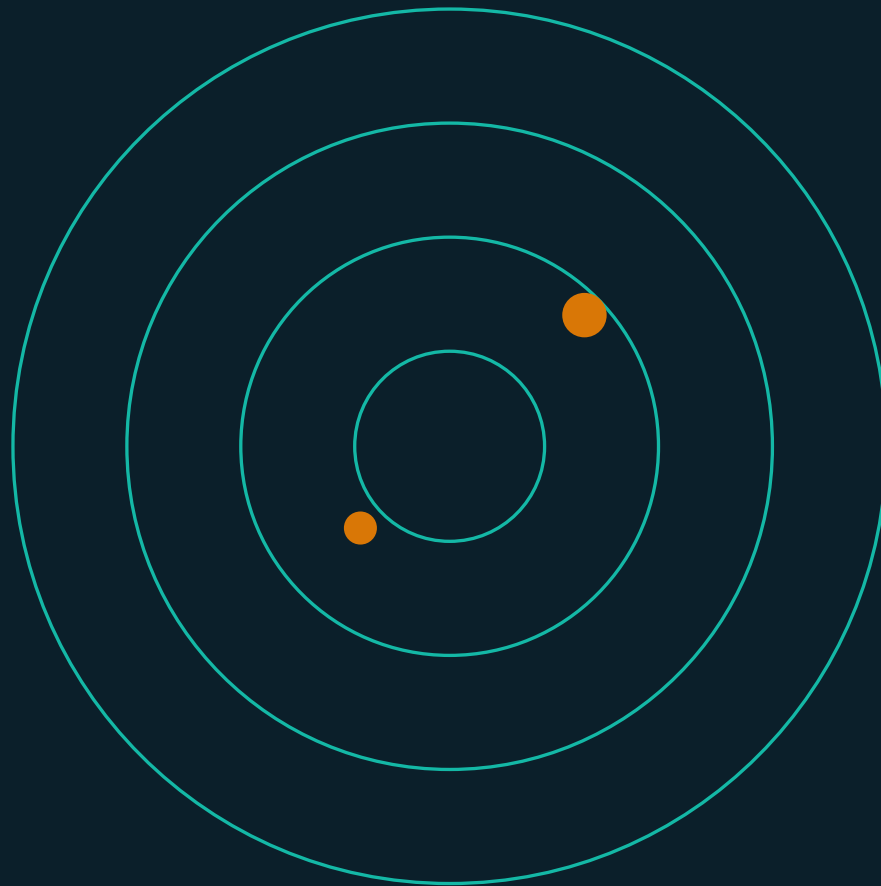


Anomaly Detection + LLM Summary

using DEFINE Foundry: Joensuu scenario, SAPIENT
detection reports



1

The data: two patterns stood out

1,234

detection reports

9

sensors

29

object IDs

PATTERN 1

0.12

One camera, constant low confidence

FI-CIV-CAM-SAVONVOIMA-GATE: 34 detections across 5 objects, every one at 0.12. Other sensors average 0.43–0.75, so this looks like a sensor-level issue.

PATTERN 2

242 s

Long gap, then reacquisition

A-02-SWM-W2_RAILWAY (HEINA-02) went unseen for ~242 s; the next-largest gaps were ~18 s. Last seen at 239.4 m, reacquired at 208.0 m.

2

Simple, explainable detector: IQR rules

Low confidence

confidence < 0.270 ($Q1 - 1.5 \times IQR$)

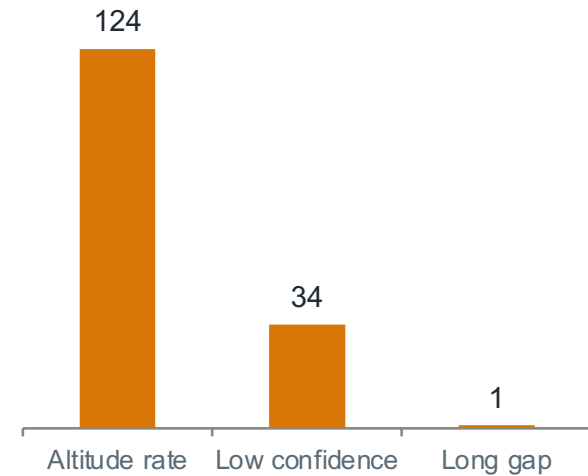
Long detection gap

time since previous report > 28.4 s ($Q3 + 1.5 \times IQR$), per object $\times$ sensor

High altitude change rate

$|\Delta \text{altitude}| \div \Delta t > 0.230 \text{ m/s}$ ($Q3 + 1.5 \times IQR$). A rate, so long gaps do not inflate it

Flags by reason



159 of 1,234 messages flagged (12.9%). Thresholds are computed from the data, and each flag keeps its reason.

2

What it catches, and where it is weak

Flag	Object	Sensor	Evidence
Low confidence	CAM-A-01_W1_DECOY	FI-CIV-CAM-SAVONVOIMA-GATE	0.12 (threshold 0.27)
Long gap	A-02-SWM-W2_RAILWAY	FI-MIL-MRAD-HEINA-02	242 s gap (threshold 28.4 s)
High altitude rate	A-02-SWM-W2_RAILWAY	FI-MIL-MRAD-HEINA-02	5.0 m in 11 s = 0.45 m/s (threshold 0.23)

Works

- Every flag carries a plain-language reason
- Rate normalisation: the 31.4 m jump over 242 s is only 0.13 m/s, so it is a gap flag, not an altitude event
- Grouping by object × sensor keeps comparisons like-for-like

Weak spots

- The 34 low-confidence flags are one constant 0.12: a sensor issue, not 34 events
- Top altitude-rate flags are all ≤ 0.45 m/s within 03:05:20–03:06:00: statistical, not necessarily operational

3

LLM situation summary (Gemini)

Input

- 40 detections from 03:05:20–03:06:00 UTC, every 3rd row kept: 12 rows
- Columns: timestamp, sensor, object_id, altitude, confidence
- Prompt: 2–4 sentences, use only the given detections, do not invent object types, locations, intent or threats

Model output

"Between 03:05:20 and 03:05:54 UTC, three sensors recorded twelve detections corresponding to seven unique object IDs (designations under the A-01 and A-02 series referencing RAILWAY, PLANT, and BRIDGE targets). Observation confidence levels across all detections range between 0.63 and 0.83, with overall recorded altitudes ranging from 213.0 down to 153.0. A consistent pattern across every object detected multiple times (such as A-01-SWM-W2_PLANT, A-02-SWM-W2_PLANT, A-02-SWM-W2_RAILWAY, and A-02-SWM-W2_BRIDGE) is a steady decrease in altitude over successive timestamps."

3

Critical evaluation of the LLM summary

Good aspects

- Correct observation period: 03:05:20 to 03:05:54 UTC
- Correctly reported that multiple sensors were involved
- Correct confidence range (0.63 to 0.83) and altitude range (153.0 m to 213.0 m)
- Avoided unsupported claims about threats, intentions or object types

Limitations and possible hallucination

- “Every object detected multiple times shows a steady decrease” is slightly too strong: e.g. A-02-SWM-W2_RAILWAY 208.0 → 203.0 m, A-02-SWM-W2_PLANT 180.0 → 172.6 m, but a short window does not prove the trend for all objects
- “Targets” is an interpretation: the data only has object_id values and sensor measurements

4

Reflection

AI helped most

- Saved time exploring data and debugging code
- Explained the SAPIENT message structure
- Helped design a simple anomaly detection method

AI needed checking

- Some suggestions were too general and missed dataset details
- Summary over-generalised the altitude trend from a short window
- Used “targets”, but the data only has object IDs and measurements
- Outputs must be reviewed before any decision-making

Production risks

- Reliability and false alarms: normal situations may be flagged
- Missing context: historical data, better tracking, domain knowledge
- Summary needs a strong link to verified sensor data
- Human validation is essential in safety-critical use